Operators in JAVA

🔹 Operators in Java

Today I explored Operators in Java, which are one of the most fundamental building blocks of programming.

An operator is a special symbol that tells the Java compiler to perform a specific operation on variables or values (called operands).

Example:

int sum = 10 + 5;

Here:
+ → Operator
10 and 5 → Operands

The operator performs the addition operation.

Operators allow us to perform:

- Mathematical calculations
- Comparisons between values
- Logical decision making
- Assigning values to variables
- Bit-level operations

1. Arithmetic Operators

Arithmetic operators are used for basic mathematical operations.

Operator	Meaning	Example
+	Addition	a + b
-	Subtraction	a - b
*	Multiplication	a * b
/	Division	a / b
%	Modulus (remainder)	a % b

Example:

```
int a = 20;
int b = 6;

System.out.println(a + b); // 26
System.out.println(a - b); // 14
System.out.println(a * b); // 120
System.out.println(a / b); // 3
System.out.println(a % b); // 2
```

📌 **Important Concept**

When dividing integers in Java:

20 / 6 = 3

Java removes the decimal part because both operands are integers.

2. Unary Operators

Unary operators work on only one operand.

Operator	Meaning
+	Positive value
-	Negative value
++	Increment
--	Decrement
!	Logical NOT

Example:

```
int x = 5;

x++;   // x becomes 6
x--;   // x becomes 5
```

Pre-increment vs Post-increment

```
int a = 5;

System.out.println(++a); // 6 (increment first)
System.out.println(a++); // 6 (print then increment)
```

After this, a becomes 7.

3. Relational (Comparison) Operators

Relational operators compare two values and return a boolean result (true or false).

Operator	Meaning
==	Equal to
!=	Not equal
>	Greater than
<	Less than
>=	Greater or equal
<=	Less or equal

Example:

```
int x = 15;
int y = 20;

System.out.println(x > y);  // false
System.out.println(x < y);  // true
System.out.println(x == y); // false
```

📌 These operators are mainly used in:

- if statements
- loops
- decision making

4. Logical Operators

Logical operators combine multiple conditions.

Operator	Meaning
&&	Logical AND
!	Logical NOT

Example:

```
int age = 22;

System.out.println(age > 18 && age < 30);
```

Explanation:

Condition 1 → age > 18
Condition 2 → age < 30

Both must be true for && to return true.

Example of OR:

```
System.out.println(age < 18 || age > 60)
```

5. Assignment Operators

Assignment operators assign values to variables.

Basic operator:

=

Example:

```
int x = 10;
```

Compound assignment operators:

Operator	Example	Meaning
+=	x += 5	x = x + 5
-=	x -= 5	x = x - 5
*=	x *= 5	x = x * 5
/=	x /= 5	x = x / 5

Example:

```
int num = 10;

num += 5;  // 15
num *= 2;  // 30
```

6.Bitwise Operators (Advanced)

Bitwise operators work on binary numbers (bits).

Operator	Meaning
&	Bitwise AND
|	Bitwise OR
^	XOR
~	Complement
<<	Left shift
>>	Right shift

Example:

```
int a = 5;
int b = 3;

System.out.println(a & b);
```

These operators are used in low-level programming and performance optimization.

7. Ternary Operator

A short form of if-else.

Operator:

? :

Example:

```
int a = 10;
int b = 20;

int max = (a > b) ? a : b;
```

Meaning:
If condition is true → first value
Else → second value

📌 Key Takeaway

Operators are extremely important in Java because they allow us to:

- Perform calculations
- Compare values
- Combine conditions
- Modify variables
- Work with binary data

Mastering operators is essential because they are used in almost every Java program.